

SAFETY DATA SHEET**SECTION 1 : CHEMICAL & COMPANY IDENTIFICATION**

PRODUCT NAME : Amoil Celplex EP 1, EP 2
 SUPPLIER : Amoil (Malaysia) Sdn.Bhd.
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MATERIAL USES:

Amoil Celplex EP 1 & EP 2 greases are high performance, long life, EP greases designed for trouble-free lubrication of a wide range of automotive and industrial equipment.

SECTION 2 : COMPOSITION & INGREDIENTS

<u>COMPOSITION</u>	<u>CAS#</u>	<u>%(W/W)</u>
Mixture of AMOIL ULTRA PURE 21 st CENTURY and/or solvent-refined base oil (petroleum) and other proprietary, non-hazardous additives.	Mixture	100

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 3 : HAZARDS IDENTIFICATION**POTENTIAL HEALTH EFFECTS:**

Prolonged or repeated contact may cause skin irritation, defatting, drying and dermatitis. Not expected to cause more than slight skin or eye irritation. With its relatively low vapour pressure, this product is not expected to be inhaled in any appreciable quantity at ambient conditions. If heated to high temperatures or subjected to mechanical actions which produce vapours or mists, inhalation may cause respiratory tract irritation. Ingestion may produce a laxative effect. For more information refer to Section 11 of this MSDS.

SECTION 4 : FIRST AID MEASURES**EYE CONTACT:**

IMMEDIATELY flush eyes with running water for at least 15 minutes, keeping eyelids open. Seek medical attention.

SKIN CONTACT:

Remove contaminated clothing - launder before reuse. Wash gently and thoroughly the contaminated skin with running water and non-abrasive soap. High pressure grease gun is capable of injecting grease through the skin. Grease gun injuries require immediate physician assessment. Seek medical attention.

INHALATION:

Evacuate the victim to a safe area as soon as possible. If the victim is not breathing, perform artificial respiration. Allow the victim to rest in a well ventilated area. Seek medical attention.

INGESTION:

DO NOT induce vomiting because of danger of aspirating liquid into lungs. Seek medical attention.

NOTE TO PHYSICIAN:

Not available

SECTION 5 : FIRE FIGHTING MEASURES

FLAMMABILITY:

May be combustible at high temperature.

FLAMMABLE LIMITS:

Not available.

FLASH POINTS Open Cup:

290°C (554°F) (Cleveland)

AUTO-IGNITION TEMPERATURE:

Fire Point: 300°C (572°F)

FIRE HAZARDS IN PRESENCE OF VARIOUS SUBSTANCES:

Low fire hazard. This material must be heated before ignition will occur.

EXPLOSION HAZARDS IN PRESENCE OF VARIOUS SUBSTANCES:

Do not cut, weld, heat, drill or pressurize empty container. Containers may explode in heat of fire.

PRODUCTS OF COMBUSTION:

Carbon oxides (CO, CO₂), nitrogen oxides (NO_x), sulphur compounds (H₂S), calcium oxides (CaO_x), lithium compounds, acrolein, aldehydes, antimony oxides, hydrocarbons, smoke and irritating vapours as products of incomplete combustion.

FIRE FIGHTING MEDIA AND INSTRUCTIONS:

NAERG96, GUIDE 171, Substances (low to moderate hazard). If tank, rail car or tank truck is involved in a fire, ISOLATE for 800 meters (0.5 mile) in all directions; also, consider initial evacuation for 800 meters (0.5 mile) in all directions. Shut off fuel to fire if it is possible to do so without hazard. If this is impossible, withdraw from area and let fire burn out under controlled conditions. Withdraw immediately in case of rising sound from venting safety device or any discolouration of tank due to fire. Cool containing vessels with water spray in order to prevent pressure build-up, autoignition or explosion.

SMALL FIRE:

Use DRY chemicals, foam, water spray or CO₂.

LARGE FIRE:

Use water spray, fog or foam. For small outdoor fires, portable fire extinguishers may be used, and self contained breathing apparatus (SCBA) may not be required. For all

indoor fires and any significant outdoor fires, SCBA is required. Respiratory and eye protection are required for fire fighting personnel.

SECTION 6 : ACCIDENTAL RELEASE MEASURES
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MATERIAL RELEASE OR SPILL:

Consult current National Emergency Response Guide Book (NAERG) for appropriate spill measures if necessary. Extinguish all ignition sources. Stop leak if safe to do so. Dike spilled material. Use appropriate inert absorbent material to absorb spilled product. Collect used absorbent for later disposal. Avoid contact with spilled material. Avoid contaminating sewers, streams, rivers and other water courses with spilled material. Notify appropriate authorities immediately.

SECTION 7 : HANDLING AND STORAGE

HANDLING:

Avoid contact with any sources of ignition, flames, heat, and sparks. Avoid eye contact. Avoid skin contact. Avoid inhalation of product vapours or mists. Empty containers may contain product residue. Do not pressurize, cut, heat, or weld empty containers. Do not reuse containers without commercial cleaning and/or reconditioning. Personnel who handle this material should practice good personal hygiene during and after handling to help prevent accidental ingestion of this product. Properly dispose of contaminated leather articles including shoes that cannot be decontaminated.

STORAGE:

Store away from incompatible and reactive materials (See section 5 and 10). Keep container tightly closed. Store in dry, cool, well-ventilated area.

SECTION 8 : EXPOSURE CONTROLS / PERSONAL PROTECTION
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ENGINEERING CONTROLS:

For normal application, special ventilation is not necessary. If user's operations generate vapours or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit. Make-up air should always be supplied to balance air removed by exhaust ventilation. Ensure that eyewash station and safety shower are close to work-station.

PERSONAL PROTECTION

The selection of personal protective equipment varies, depending upon conditions of use.

EYES:

Eye protection (i.e., safety glasses, safety goggles and/or face shield) should be determined based on conditions of use. If product is used in an application where splashing may occur, the use of safety goggles and/or a face shield should be considered.

BODY:

Wear appropriate clothing to prevent skin contact. As a minimum long sleeves and trousers should be worn.

RESPIRATORY:

Where concentrations in air may exceed the occupational exposure limits given in Section 2 (and those applicable to your are) and where engineering, work practices or other means of exposure reduction are not adequate, NIOSH approved respirators may be necessary to prevent overexposure by inhalation.

HANDS:

Wear appropriate chemically protective gloves. When handling hot product ensure gloves are heat resistant and insulated.

FEET:

Wear appropriate footwear to prevent product from coming in contact with feet and skin.

EXPOSURE LIMITS

This product is not expected to form a mist based on its properties and expected use.

SECTION 9 : PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE	:	Stringy smooth semi-solid.
COLOUR	:	Green.
ODOUR	:	Mild grease like.
ODOUR THRESHOLD	:	Not available.
BOILING POINT	:	Not available.
SPECIFIC GRAVITY	:	0.8893 kg/L @ 15°C (59°F).
VAPOR DENSITY	:	Not available.
VAPOR PRESSURE	:	Negligible at ambient temperature and pressure.
VOLATILITY	:	Non-volatile.
VISCOSITY	:	219.9 cSt @ 40°C (104°F), 17.9 cSt @ 100°C (212°F)
POUR POINT	:	-12°C (10°F)
SOFTENING POINT	:	Not applicable.
DROPPING POINT	:	EP 1: 307°C (585°F) EP 2: 302°C (576°F)
PENETRATION	:	333 (EP 1), 271 (EP 2), 60 strokes.
OIL / WATER DIST. COEFF.	:	Not available.
IONICITY (IN WATER)	:	Not available.
DISPERSION PROPERTIES	:	Not available.
SOLUBILITY	:	Insoluble in water.

SECTION 10 : STABILITY AND REACTIVITY
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CORROSIVITY:

Not corrosive to copper or steel.

STABILITY:

The product is stable under normal handling and storage conditions.

HAZARDOUS POLYMERIZATION:

Will not occur under normal working conditions.

INCOMPATIBLE SUBSTANCES / CONDITIONS TO AVOID:

Reactive with oxidizing agents, acids, alkalis, salts, maleic anhydride, phosphorus and reducing agents.

DECOMPOSITION PRODUCTS:

May release CO_x, NO_x, SO_x, H₂S, PO_x, diphenylamine, alkenes, lithium compounds, smoke and irritating vapours when heated to decomposition.

SECTION 11 : TOXICOLOGICAL INFORMATION

ROUTES OF ENTRY:

Skin contact, eye contact, inhalation and ingestion.

ACUTE LETHALITY:

Acute toxicity information is not available for the product as a whole, therefore, data for some of the ingredients is provided below:

Acute oral toxicity (LD50): >5000 mg/kg (rat).

Acute dermal toxicity (LD50): >2000 mg/kg (rabbit).

CHRONIC OR OTHER TOXIC EFFECTS

DERMAL ROUTE:

Prolonged or repeated contact may defat and dry skin, and cause dermatitis. Short-term exposure is expected to cause only slight irritation, if any.

INHALATION ROUTE:

With its relatively low vapour pressure, this product is not expected to be inhaled in any appreciable quantity at ambient conditions. If heated to high temperatures or subjected to mechanical actions which produce vapours or mists, inhalation may cause respiratory tract irritation.

ORAL ROUTE:

Ingestion of this product may lead to aspiration of the liquid, especially if vomiting occurs. This may result in chemical pneumonitis (inflammation of the lungs) and/or pulmonary edema (an accumulation of fluid in the lungs). May produce a laxative effect.

EYE IRRITATION / INFLAMMATION:

Short-term exposure is expected to cause only slight irritation, if any.

IMMUNOTOXICITY:

Not available.

SKIN SENSITIZATION:

Contact with this product is not expected to cause skin sensitization, based upon the available data and the known hazards of the components.

RESPIRATORY TRACT SENSITIZATION:

Contact with this product is not expected to cause respiratory tract sensitization, based upon the available data and the known hazards of the components.

MUTAGENIC:

This product is not known to contain any components at $\geq 0.1\%$ that have been shown to cause mutagenicity. Therefore, based upon the available data and the known hazards of the components, this product is not expected to be a mutagen.

REPRODUCTIVE TOXICITY:

This product is not known to contain any components at $\geq 0.1\%$ that have been shown to cause reproductive toxicity. Therefore, based upon the available data and the known hazards of the components, this product is not expected to be a reproductive toxin.

TERATOGENICITY / EMBRYOTOXICITY:

This product is not known to contain any components at $\geq 0.1\%$ that have been shown to cause teratogenicity and/or embryotoxicity. Therefore, based upon the available data and the known hazards of the components, this product is not expected to be a teratogen/embryotoxin.

CARCINOGENICITY (ACGIH):

This product is not known to contain any chemicals at reportable quantities that are listed as Group A1 or A2 carcinogens by ACGIH.

CARCINOGENICITY (IARC):

This product is not containing any chemicals at reportable quantities that are listed as Group 1, 2A, or 2B carcinogens by IARC.

CARCINOGENICITY (NTP):

This product is not known to contain any chemicals at reportable quantities that are listed as carcinogens by NTP.

CARCINOGENICITY (IRIS):

This product is not known to contain any chemicals at reportable quantities that are listed as carcinogens by IRIS.

CARCINOGENICITY (OSHA):

This product is not known to contain any chemicals at reportable quantities that are listed as carcinogens by OSHA.

SECTION 12 : ECOLOGICAL INFORMATION

ENVIRONMENTAL FATE	: Not available.
BOD5 AND COD	: Not available.
ADDITIONAL REMARKS	: No additional remark.
PERSISTANCE/BIOACCUMULATION POTENTIAL	: Not available.
PRODUCTS OF BIODEGRADATION	: Not available.

SECTION 13 : DISPOSAL CONSIDERATIONS

WASTE DISPOSAL:

Spent/ used/ waste oil may meet the requirements of a hazardous waste. Consult your local or regional authorities. Ensure that waste management processes are in compliance with government requirements and local disposal regulations.

SECTION 14 : TRANSPORT INFORMATION**SPECIAL PROVISIONS FOR TRANSPORT:**

Not applicable.

SECTION 15: REGULATORY INFORMATION**OTHER REGULATIONS:**

This product is acceptable for use under the provisions of WHMIS-CPR. All components of this formulation are listed on the CEPA-DSL (Domestic Substances List).

All components of this formulation are listed on the US EPA-TSCA Inventory.

All components of this formulation are listed on EINECS or are exempt.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all of the information required by the CPR.

SECTION 16 : OTHER INFORMATION

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All material may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

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